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Backward Analysis of the Total Least Squares and its Randomized Algorithms

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Question: Why are we interested in the backward error analysis of the TLS?

Development	least squares (LS)	total least squares (TLS)
Formulation	$\min_x \ Ax - b\ _2$	$\min_{\Delta A, \Delta b, x} \ [\Delta A, \Delta b]\ _F, \text{ s.t. } (A + \Delta A)x = b + \Delta b$
Normal equation	$A^\top(b - Ax) = 0$	$(A^\top A - \sigma'_{n+1}{}^2 I)x = A^\top b$
Direct method / Closed form	QR: $x = R^\dagger Qb$	SVD: $x = -\frac{v'_{n+1}[1:n]^\top}{v'_{n+1}[n+1]}, B \equiv [A, b] = U\Sigma V^\top.$
【most recent】 Solvers and techniques	Sketching-and-solve; Iterative refinement; Randomized preconditioning;...	 Relatively underdeveloped
Null-space estimation : given $A \in \mathbb{R}^{m \times n} (m \gg n)$, consider $\min_{V^\top V = I_k} \ AV\ _F .$		

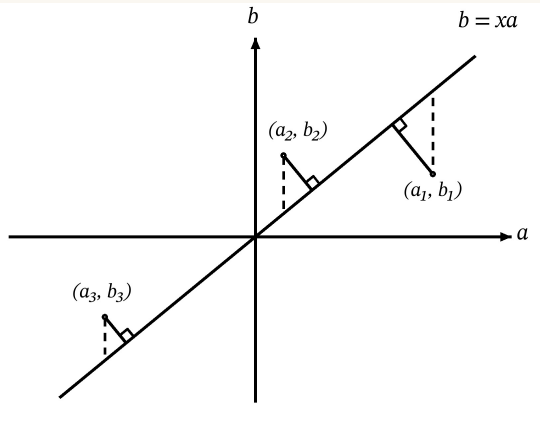


Fig: Least squares (--) VS Total least squares (—)

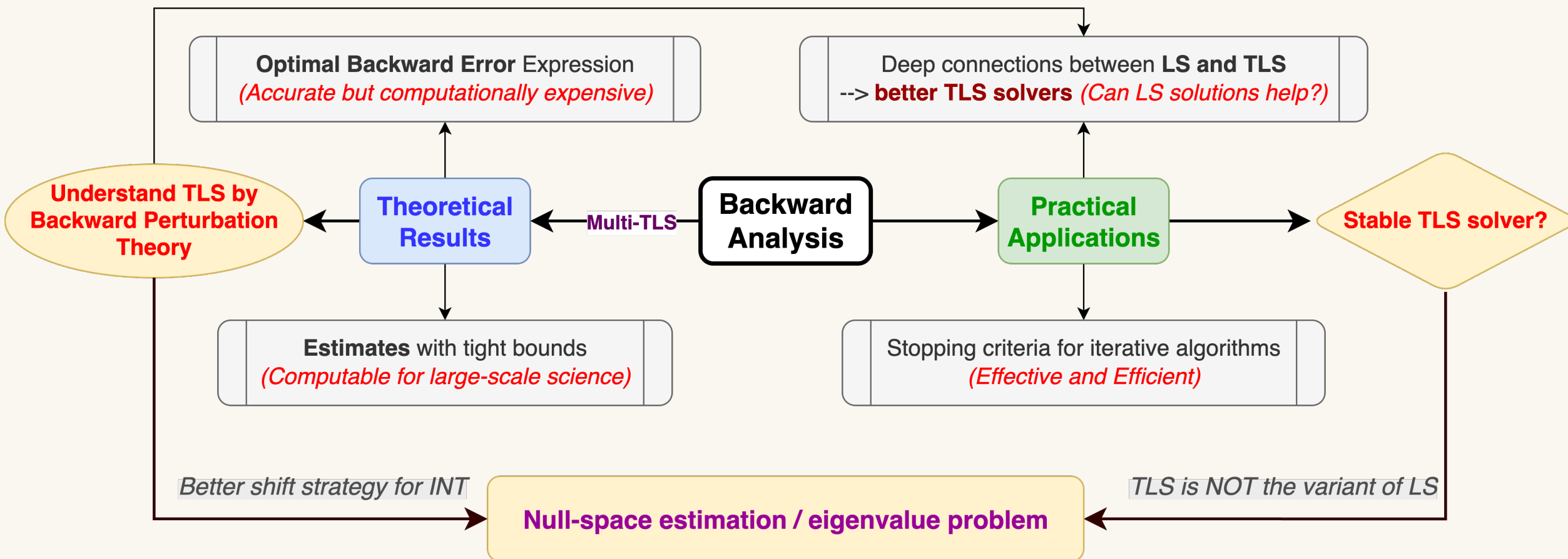
• **Research direction** : TLS problem can leverage backward-stable LS solvers, but such techniques **do not** extend directly to the general setting, due to fundamental differences in the **structured normal equations, backward-error characterization, and others**.

• **Shan, Jiali**, and Yimin Wei. "Optimal backward error of a total least squares and its randomized algorithms." *SIAM Journal on Matrix Analysis and Applications* 46.3 (2025): 2116-2139.

• **Research approach** : Move beyond the conventional **one-way view of TLS as an application** of null-space estimation. By exploiting the **TLS structure** and **optimal backward error**, aim to overcome longstanding bottlenecks in classical eigensolvers such as **LOBPCG**.

• **Shan, Jiali**, and Yimin Wei. "Efficient estimate for the optimal backward error of the multidimensional total least squares." *SIAM Journal on Matrix Analysis and Applications* 47.2 (2026): 649-672.

Outline



Backward Perturbation for the multidimensional TLS (mTLS)

(mTLS) Problem formulation: $\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F$, subject to $(A + \Delta A)X = B + \Delta B$, where $A \in \mathbb{R}^{m \times n}$, $m \gg n$.

Given a computed solution $\tilde{X} \in \mathbb{R}^{m \times n}$, the **θ -weighted backward error** of the mTLS problem is defined as

$$\eta_{\theta}(\tilde{X}) = \min \{ \|[E, \theta F]\|_F : \tilde{X} \text{ is the true solution to a nearby mTLS system } (A + E)X \approx B + F \}, \text{ i.e.,}$$

$$\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F, \text{ subject to } (A + E + \Delta A)\tilde{X} = B + F + \Delta B.$$

GOALS: ① *Theoretically* establish the backward perturbation theory for the general mTLS problem;

→ give the explicit expression of $\eta_{\theta}(\tilde{X})$.

② *Practically* develop efficient algorithms for the computation of $\eta_{\theta}(\tilde{X})$.

→ derive estimates with tight theoretical bounds and exploit randomization in the algorithm design.

Backward Perturbation Theory – Related works

Least Squares (LS) Problem —

• **Single right-hand side** : $\min_x \|Ax - b\|_2$, where $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$.

-- Given a nonzero $\hat{x} \in \mathbb{R}^n$ with $\hat{r} = b - A\hat{x}$, the following two sets are identical (*Walden, Karlson and Sun, 1995*),

$$\mathcal{E} = \{E \in \mathbb{R}^{m \times n} : \|b - (A + E)\hat{x}\| = \min_{x \in \mathbb{R}^n} \|b - (A + E)x\|\}, \mathcal{M} = \{\hat{r}\hat{x}^\dagger - vv^\dagger(A + \hat{r}\hat{x}^\dagger) + (I - vv^\dagger)Z(I - \hat{x}\hat{x}^\dagger) : v \in \mathbb{R}^m, Z \in \mathbb{R}^{m \times n}\}.$$

-- Define $\lambda = \lambda_{\min}\left(AA^\top - \frac{\hat{r}\hat{r}^\top}{1 + \|\hat{x}\|^2}\right)$ and $\mathcal{G} = \{[E, f] : \|(b + f) - (A + E)\hat{x}\| = \min_{x \in \mathbb{R}^n} \|(b + f) - (A + E)x\|\}$, then

$$\mu(\hat{x}) = \min_{[E, f] \in \mathcal{G}} \|[E, f]\|_F = \begin{cases} \frac{\|\hat{r}\|}{\|\hat{x}\|}, & \text{if } \lambda \geq 0, \\ \left(\frac{\|\hat{r}\|^2}{\|\hat{x}\|^2} + \lambda\right)^{\frac{1}{2}}, & \text{if } \lambda < 0. \end{cases} \quad (\text{Walden and Karlson, 1997}) \rightarrow \text{Computationally expensive and unstable}$$

-- Stable alternative: $\mu(\hat{x}) = \min\left\{\frac{\|\hat{r}\|}{\|\hat{x}\|}, \sigma_{\min}[A, B]\right\}, B = \frac{\|\hat{r}\|}{\|\hat{x}\|}(I - \hat{r}\hat{r}^\dagger)$ (*Higham, 2002*)

Efficient estimate: $\tilde{\mu}(\hat{x}) = \left\|(\|\hat{x}\|^2 AA^\top + \|\hat{r}\|^2 I)^{-\frac{1}{2}} A^\top \hat{r}\right\|$ (*Walden and Karlson, 1997; Gu, 1998; Gratton, 2012; Grcar et al., 2008*)

• **Multiple right-hand side** (*Hallman, 2020*)

Backward Perturbation Theory – Related works

Total Least Squares (TLS) Problem —

• **Single right-hand side** : (Scaled TLS) $\min_{\Delta A, \Delta b} \|[\Delta A, \gamma \Delta b]\|_F$, subject to $(A + \Delta A)x = b + \Delta b$, where $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m, m \gg n$.

-- The θ -weighted backward error for a computed solution $\tilde{x} \in \mathbb{R}^n$ is

$$\min_{E, f} \| [E, \theta f] \|_F, \text{ subject to } \tilde{x} \text{ solves a nearby scaled TLS problem } (A + E)x \approx b + f.$$

-- Denote $\tilde{r} = b - A\tilde{x}$ and $M = A(I - \tilde{x}\tilde{x}^\top)A^\top - \frac{\theta^2 \tilde{r}\tilde{r}^\top}{1 + \theta^2 \|\tilde{x}\|^2} + \frac{\theta^2 (A\tilde{x} + \gamma^2 \|\tilde{x}\|^2 b)(A\tilde{x} + \gamma^2 \|\tilde{x}\|^2 b)^\top}{\theta^2 \|\tilde{x}\|^2 + \gamma^4 \|\tilde{x}\|^4}$, then (*Chang et al., 2009*)

$$\min_{E, f} \| [E, \theta f] \|_F^2 = \begin{cases} \frac{\theta^2 \|\tilde{r}\|^2}{1 + \theta^2 \|\tilde{x}\|^2}, & \text{if } \lambda_{\min}(M) \geq 0, \\ \frac{\theta^2 \|\tilde{r}\|^2}{1 + \theta^2 \|\tilde{x}\|^2} + \lambda_{\min}(M), & \text{if } \lambda_{\min}(M) < 0. \end{cases} \quad \rightarrow \text{Though explicit, not easy to understand by intuition}$$

• **Multiple right-hand side** (*Shan and Wei, 2025*)

→ Single right-hand side is naturally the special case : Accurate, explicit and intuitively right! Deeply connected with LS.

mTLS problem from an optimization perspective

The problem $\min_{\Delta A, \Delta b} \|\Delta A, \Delta b\|_F$, subject to $(A + \Delta A)x = b + \Delta b$ is equivalent to the $\operatorname{argmin}_y f(y) = \frac{\|b - Ay\|^2}{1 + \|y\|^2}$. → mTLS?

Theorem 1: $X \in \mathbb{R}^{n \times d}$ is the true solution to the mTLS problem

$$\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F, \text{ subject to } (A + \Delta A)X = B + \Delta B,$$

if it satisfies $A^T(B - AX) = -X(X^T X + I)^{-1}(B - AX)^T(B - AX)$, or $(A + BX^T)^T(B - AX) = 0$.

Important takeaway: We can characterize the perturbations to the mTLS systems in an elegant way —

$$\mathcal{F}_{\text{mTLS}+} = \{[E, F] \in \mathbb{R}^{m \times (n+d)} : ((A + E) + (B + F)X^T)^T((B + F) - (A + E)X^T) = 0\}$$

→ Why the relaxed feasible set could be useful? We shall expect the computed solution as a reasonable approximation to the true one.

Computed solution $\bar{X}$ with full column-rank

Theorem 2 (characterization of the perturbation to A when the right-hand side is fixed): Given $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times d}$, $\bar{X} \in \mathbb{R}^{n \times d}$ with full column-rank, then the following two sets are identical,

$$\begin{aligned}\mathcal{E}_{\text{mTLS}} &= \{E \in \mathbb{R}^{m \times n}: ((A + E) + B\bar{X}^\top)^\top [B - (A + E)\bar{X}] = 0\}, \\ \mathcal{M}_{\text{mTLS}} &= \{-YY^\dagger(\mathbf{A} + \mathbf{B}\bar{X}^\top) + (I - YY^\dagger)[\hat{R}\bar{X}^\dagger + Z(I - \bar{X}\bar{X}^\dagger)]: Y \in \mathbb{R}^{m \times d}, Z \in \mathbb{R}^{m \times n}\}, \text{ with } \hat{R} = B - A\bar{X}.\end{aligned}$$

Comparison with mLS — Deep connections with Least Squares!

Given $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times d}$, $\hat{X}_{\text{LS}} \in \mathbb{R}^{n \times d}$ with full column-rank, then the following two sets are identical (*Sun, 1996*),

$$\begin{aligned}\mathcal{E}_{\text{LS}} &= \{E \in \mathbb{R}^{m \times n}: \|(A + E)\hat{X}_{\text{LS}} - B\|_F = \min_{X \in \mathbb{R}^{n \times d}} \|(A + E)X - B\|_F\}, \\ \mathcal{M}_{\text{LS}} &= \{-YY^\dagger \mathbf{A} + (I - YY^\dagger)[\hat{R}_{\text{LS}}\hat{X}_{\text{LS}}^\dagger + Z(I - \hat{X}_{\text{LS}}\hat{X}_{\text{LS}}^\dagger)]: Y \in \mathbb{R}^{m \times d}, Z \in \mathbb{R}^{m \times n}\}, \text{ with } \hat{R}_{\text{LS}} = B - A\hat{X}_{\text{LS}}.\end{aligned}$$

Theoretical explicit optimal backward error for mTLS

Recall that $\eta_\theta(\tilde{X}) = \min \{ \|[E, \theta F]\|_F : \tilde{X} \text{ is the true solution to a nearby mTLS system } (A + E)X \approx B + F \}, \text{ i.e.,}$

$$\min_{\Delta A, \Delta B} \|[\Delta A, \Delta B]\|_F, \text{ subject to } (A + E + \Delta A)\tilde{X} = B + F + \Delta B.$$

Theorem 3 (optimal backward error with all data perturbed): Let $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{m \times d}, \tilde{X} \in \mathbb{R}^{n \times d}$ with full column-rank. Define

$$\tilde{A} = (A + B\tilde{X}^\top)(I + \theta^{-2}\tilde{X}\tilde{X}^\top)^{-\frac{1}{2}} \text{ and } \tilde{R} = \theta(B - A\tilde{X})(I + \theta^2\tilde{X}^\top\tilde{X})^{-\frac{1}{2}}, \text{ where } \theta \text{ is the positive weighting parameter. Write } \lambda_1^{(\theta)} \leq \lambda_2^{(\theta)} \leq$$

$\dots \leq \lambda_m^{(\theta)}$ as the eigenvalues of the matrix $\tilde{A}\tilde{A}^\top - \tilde{R}\tilde{R}^\top$. Then, the optimal backward error is $\eta_\theta(\tilde{X}) = \left[\|\tilde{R}\|_F^2 + \lambda_*^{(\theta)} \right]^{\frac{1}{2}}$, where

$$\lambda_*^{(\theta)} = \begin{cases} 0, & \text{if } 0 \leq \lambda_1^{(\theta)}, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \dots + \lambda_s^{(\theta)}, & \text{if } \lambda_s^{(\theta)} < 0 \leq \lambda_{s+1}^{(\theta)} \text{ with } s < d, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \dots + \lambda_d^{(\theta)}, & \text{if } \lambda_d^{(\theta)} < 0. \end{cases}$$

- For the rank-deficient case & limited data-perturbation case, please go to our original paper.

Practical computation (Single right-hand side)

Corollary 4 (Optimal backward error for a TLS): By setting the parameter $d = 1$, we have the optimal backward error for the TLS system with a single right-hand side as

$$\tau_{\theta}(\tilde{x}) = \min_{E, f} \|[E, \theta f]\|_F = \left[\|\tilde{r}\| + \min \{0, \lambda_{\min}(\tilde{A}\tilde{A}^{\top} - \tilde{r}\tilde{r}^{\top})\} \right]^{\frac{1}{2}},$$

where $\tilde{A} = (A + b\tilde{x}^{\top})(I + \theta^{-2}\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}(b - A\tilde{x})$.

For the numerical stability, it should be computed by

$$\min \{\|\tilde{r}\|, \sigma_{\min}(M)\}, \text{ with } M = \begin{pmatrix} \tilde{A}^{\top} \\ \|\tilde{r}\|(I - \tilde{r}\tilde{r}^{\dagger}) \end{pmatrix}.$$

Theorem 5 (Computable estimate for a TLS): Define

$$\tilde{\tau}_{\theta}(\tilde{x}) = \left\| (\tilde{A}^{\top}\tilde{A} + \|\tilde{r}\|^2 I)^{-\frac{1}{2}} \tilde{A}^{\top} \tilde{r} \right\|,$$

where $\tilde{A} = (A + b\tilde{x}^{\top})(I + \theta^{-2}\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}(b - A\tilde{x})$, then $\tilde{\tau}_{\theta}(\tilde{x}) \leq \tau_{\theta}(\tilde{x}) \leq \sqrt{2}\tilde{\tau}_{\theta}(\tilde{x})$.

Practical computation with Sketching techniques

Definition 6 (Sketching matrix): A random matrix $S \in \mathbb{R}^{s \times m}$ is called a sketching matrix for a matrix $A \in \mathbb{R}^{m \times n}$ with distortion $\epsilon \in (0,1)$, if

$$\forall y \in \mathbb{R}^n, (1 - \epsilon)\|Ay\| \leq \|S(Ay)\| \leq (1 + \epsilon)\|Ay\|.$$

For probabilistic sketching matrices, they satisfy the property with high probability.

Lemma 7 (Randomized preconditioning): Let S be a sketching matrix for a full-rank matrix A with distortion $\epsilon \in (0,1)$.

Consider a factorization $SA = QK$, where Q is orthogonal and K is square. Then,

$$\frac{1}{1+\epsilon} \leq \sigma_{\min}(AK^{-1}) \leq \sigma_{\max}(AK^{-1}) \leq \frac{1}{1-\epsilon}. \text{ (Woolfe et al. , 2008)}$$

Practical computation with Sketching techniques

Theorem 8 (Computable estimate for a TLS with sketching): Let $S \in \mathbb{R}^{m \times (n+1)}$ be a sketching matrix for $\text{span}([A, b])$ and $SA = QK$ as above. Write $\tilde{A} = (A + b\tilde{x}^\top)(I + \theta^{-2}\tilde{x}\tilde{x}^\top)^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^\top)^{-\frac{1}{2}}(b - A\tilde{x})$. Define $\tilde{\tau}_\theta^{sk}(\tilde{x}) = \left\| \left((S\tilde{A})^\top (S\tilde{A}) + \|\tilde{r}\|^2 I \right)^{-\frac{1}{2}} \tilde{A}^\top \tilde{r} \right\|$, then $(1 - \epsilon)\tilde{\tau}_\theta^{sk}(\tilde{x}) \leq \tau_\theta(\tilde{x}) \leq \sqrt{2}(1 + \epsilon)\tilde{\tau}_\theta^{sk}(\tilde{x})$.

Choice of sketching: sparse sign embedding $S^{d \times m} = \xi^{-\frac{1}{2}}[s_1, \dots, s_m]$, where s_i are independent random vectors with ξ nonzero entries (± 1 , both with probability $\frac{1}{2}$) placed in uniformly random positions. Choose $s \geq \text{const} \cdot \frac{n \log n}{\eta^2}$ and $\xi \geq \text{const} \cdot \frac{\log n}{\eta}$ ($s = 12n, \xi = 8$).

Algorithm 3.1 Efficient estimate for the optimal backward error of a TLS

Require: Matrix $A \in \mathbb{R}^{m \times n}$, right-hand side $b \in \mathbb{R}^m$, computed solution $\tilde{x} \in \mathbb{R}^n$, weighting parameter θ

Ensure: Sketched estimate $\tilde{\tau}_\theta^{sk}(\tilde{x})$ of the optimal backward error

- 1: Draw a sketching matrix $S \in \mathbb{R}^{s \times m}$ for $[A, b]$
- 2: $\bar{A} \leftarrow A + b\tilde{x}^\top$, $\tilde{r} \leftarrow \theta(1 + \theta^2\|\tilde{x}\|^2)^{-1/2}(b - A\tilde{x})$, $\zeta^2 \leftarrow \tilde{r}^\top \tilde{r}$
- 3: $\tilde{A} \leftarrow \bar{A} + \left(\bar{A} \cdot \left(\frac{\theta}{\sqrt{\theta^2 + \|\tilde{x}\|^2}} - 1 \right) \tilde{x} \right) \cdot \frac{\tilde{x}^\top}{\|\tilde{x}\|}$
- 4: $(\tilde{U}_{sk}, \tilde{\Sigma}_{sk}, \tilde{V}_{sk}) \leftarrow \text{svd}(S\tilde{A})$ ▷ $\tilde{\Sigma}_{sk}$ in column vector form
- 5: $\tilde{\tau}_\theta^{sk}(\tilde{x}) \leftarrow \left\| \left[\frac{1}{\sigma_i^2 + \zeta^2} \right]_{i=1}^n \odot (\tilde{V}_{sk}^\top \tilde{A}^\top \tilde{r}) \right\|$ ▷ Hadamard product $\odot$



Practical computation tests (Single right-hand side)

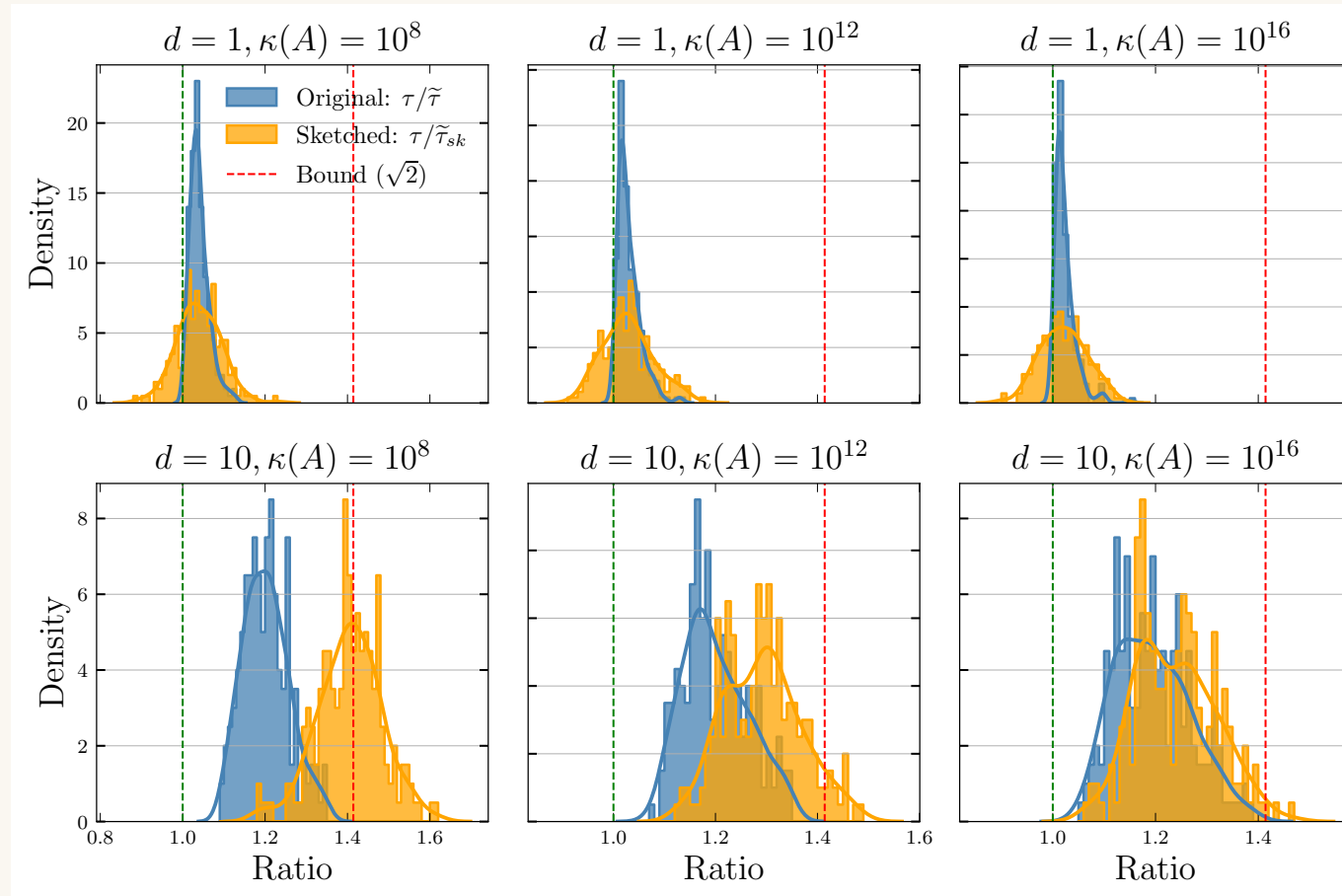


Figure: Tightness of the proposed estimations, with and without sketching techniques, for the optimal backward error of a total least squares. *Examples: fixed size $A^{3000 \times 20}$ with varying condition numbers.*

Practical computation (Multiple right-hand sides)

Theorem 9 (Computable estimate for mTLS): Denote $\tilde{A} = (A + B\tilde{X}^\top)(I + \theta^{-2}\tilde{X}\tilde{X}^\top)^{-\frac{1}{2}}$ and $\tilde{R} = \theta(B - A\tilde{X})(I + \theta^2\tilde{X}^\top\tilde{X})^{-\frac{1}{2}}$. Let $r_{\tilde{A}}$ and $r_{\tilde{R}}$ be the ranks of $\tilde{A}$ and $\tilde{R}$, with SVD decompositions $\tilde{A} = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ and $G = W\Lambda Z^\top$. Then,

(recall that) $\eta_\theta(\tilde{X}) = [||\tilde{R}||_F^2 + \text{tr}_-(\tilde{A}\tilde{A}^\top - \tilde{R}\tilde{R}^\top)]^{\frac{1}{2}} \rightarrow \eta_\theta(\tilde{X}) = \eta(\tilde{A}, \tilde{R}) = \min \left\{ ||[M, N]||_F : \tilde{A}^\top N + M^\top \tilde{R} = -\tilde{A}^\top \tilde{R} \text{ and } M^\top N = 0 \right\}.$

Thus, we have a natural bound as $v^\theta(\tilde{X}) =$, which is equivalent to

$$v^\theta(\tilde{X}) = \left[\sum_{j=1}^{r_{\tilde{R}}} \lambda_j^2 \left\| (\tilde{A}\tilde{A}^\top + \lambda_j I)^{-\frac{1}{2}} \tilde{A}^\top w_j \right\| \right]^{\frac{1}{2}}.$$

It has a tight bound of $v_\theta(\tilde{X}) \leq \eta_\theta(\tilde{X}) \leq \sqrt{2}v_\theta(\tilde{X})$.

Similarly, using the sketching techniques, $v_\theta^{sk}(\tilde{X}) = \left[\sum_{j=1}^{r_{\tilde{R}}} \lambda_j^2 \left\| ((S\tilde{A})(S\tilde{A})^\top + \lambda_j I)^{-\frac{1}{2}} \tilde{A}^\top w_j \right\| \right]^{\frac{1}{2}}$ satisfies $1 - \epsilon \leq \frac{\eta_\theta(\tilde{X})}{v_\theta^{sk}(\tilde{X})} \leq \sqrt{2}(1 + \epsilon)$.

Practical computation tests (Multiple right-hand sides)

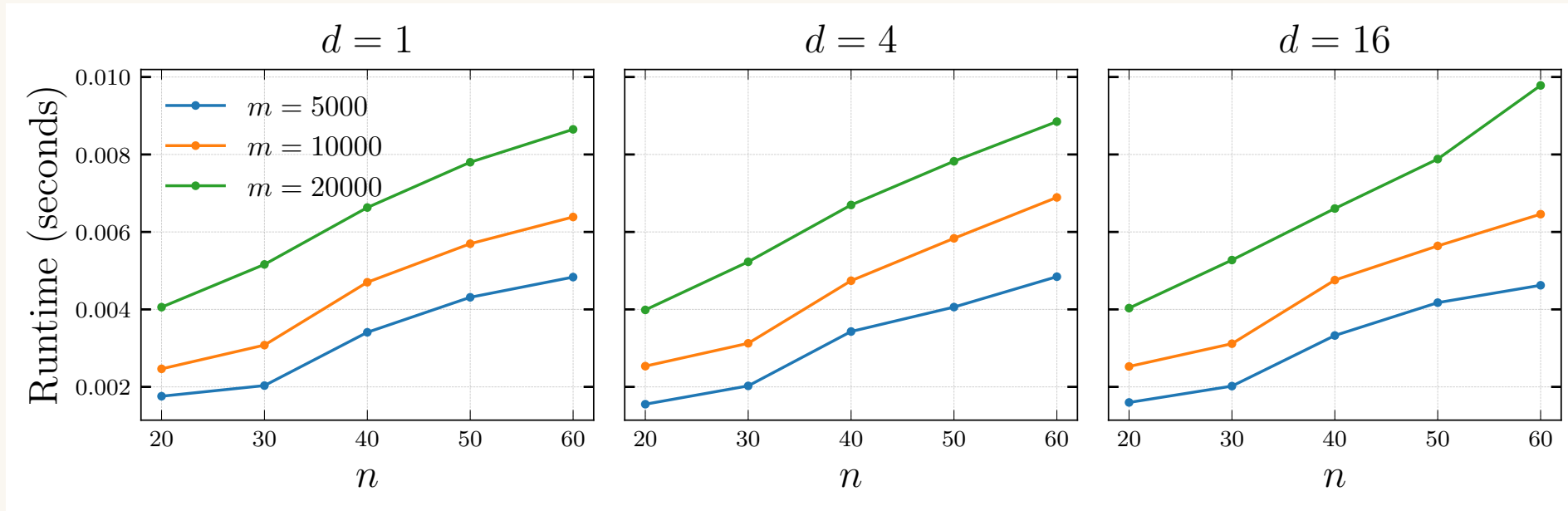


Figure: Efficiency of the sketched estimates for the optimal backward error of mTLS problems. *Examples: fixed-condition-number ($1e+6$) A with varying sizes, $m = [5000, 10000, 20000]$, $n = \text{linspace}(20, 60, 5)$, $d=[1, 4, 16]$.*

Conclusion

